

An infinite family of $5\text{-}\bar{\omega}$ -critical tournaments

(Conjecture 5.10 of Aboulker–Aubian–Charbit–Lopes at $k = 5$)

Abstract

For a tournament T the *clique number* $\bar{\omega}(T)$ is the minimum, over all orderings $\prec$ of $V(T)$, of the clique number of the backedge graph. A tournament is $k\text{-}\bar{\omega}$ -critical if $\bar{\omega}(T) = k$ and $\bar{\omega}(T - v) = k - 1$ for every vertex v . We prove that for every odd $n \geq 7$ the lexicographic substitution $AC_n[AC_n]$ of the almost-consecutive circulant $AC_n = \text{Cay}(\mathbb{Z}/n, \{1, \dots, m-1\} \cup \{m+1\})$ ($n = 2m+1$) into itself is $5\text{-}\bar{\omega}$ -critical. Hence there are infinitely many $5\text{-}\bar{\omega}$ -critical tournaments, settling Conjecture 5.10 of Aboulker, Aubian, Charbit and Lopes ([arXiv:2310.04265](#)) at $k = 5$, and the associated local–global question (Question 5.9) fails at $k = 5$. The proof reduces the deletion bound to a finite, n -independent obstruction expressed through eight “cells”, closed by elementary residue arithmetic and one interval-autocorrelation lemma.

1 Definitions and the tools used

Definition 1 (Backedge graph; tournament clique number). Let T be a tournament and $\prec$ a total order of $V(T)$. The *backedge graph* $T^\prec$ has vertex set $V(T)$ and an edge $\{u, v\}$ (with $u \prec v$) iff the arc between them is $v \rightarrow u$ (“backward”). A *backedge clique* is a clique of $T^\prec$: a set listed by $\prec$ in reverse topological order, i.e. every later vertex beats every earlier one. We set $\bar{\omega}(T) = \min_{\prec} \omega(T^\prec)$. The tournament T is $k\text{-}\bar{\omega}$ -critical if $\bar{\omega}(T) = k$ and $\bar{\omega}(T - v) = k - 1$ for all v .

Definition 2 (Substitution). For tournaments S, H , the *lexicographic substitution* $S[H]$ has vertex set $V(S) \times V(H)$ and an arc $(s, h) \rightarrow (s', h')$ iff $s \rightarrow s'$ in S (when $s \neq s'$) or $s = s'$ and $h \rightarrow h'$ in H .

Definition 3 (The base circulant). Fix $n = 2m + 1$ with $m \geq 3$ and put $g = \{1, \dots, m-1\} \cup \{m+1\} \subseteq \mathbb{Z}/n$. The *almost-consecutive circulant* is $AC_n = \text{Cay}(\mathbb{Z}/n, g)$, with arc $i \rightarrow j$ iff $(j - i) \bmod n \in g$. Since for each $k \in \{1, \dots, m\}$ exactly one of $k, n - k$ lies in g , AC_n is a tournament; it is vertex-transitive ($x \mapsto x + 1$ is an automorphism). We write $\mathbf{H} := [m+1, 2m]$ and $\mathbf{L} := [1, m]$ for the two halves of $\mathbb{Z}/n \setminus \{0\}$.

Lemma 1 (Arc-facts). *With g as above:*

- (i) for $a \in \mathbf{H}$: $a \in g \iff a = m+1$, and $(-a) \equiv n - a \in g \iff a \in [m+2, 2m]$;
- (ii) for $a \in \mathbf{L}$: $a \in g \iff a \in [1, m-1]$, and $(-a) \equiv n - a \in g \iff a = m$;
- (iii) for a, a' in one of the length- m intervals $\mathbf{H}$ or $\mathbf{L}$, the residue $(a - a') \bmod n$ lies in $[m+2, 2m] \cup [1, m-1]$, hence is in g only when $a > a'$ with $a - a' \leq m-1$ (never the wrap value $m+1$).

Proof. Direct from $g = \{1, \dots, m-1\} \cup \{m+1\}$ and $n = 2m+1$. For $a \in \mathbf{H}$, $n - a \in [1, m]$ meets g exactly on $[1, m-1]$, i.e. $a \in [m+2, 2m]$. For $a \in \mathbf{L}$, $n - a \in [m+1, 2m]$ meets g only at $m+1$, i.e. $a = m$. For (iii), two elements of a length- m interval differ by at most $m-1$ in absolute value, so the wrap value $m+1$ never occurs. $\square$

The asymmetry in Lemma 1 between the two bands— $(-a) \in g$ selects the whole interval $[m+2, 2m]$ in band H but the single point $a = m$ in band L—is genuine and is used repeatedly below.

Proposition 1 (Facts about AC_n). *For every odd $n = 2m + 1 \geq 7$:*

- (a) $\bar{\omega}(AC_n) = 3$; *in the identity order the unique backedge triangle is $\{0, m, 2m\}$.*
- (b) $\bar{\omega}(AC_n - v) = 2$ *for every v ; in particular AC_n is $3\text{-}\bar{\omega}$ -critical.*

Proof. Upper bound $\bar{\omega}(AC_n) \leq 3$ and the unique triangle. In the identity order a backedge $i < j$ means j beats i , i.e. $(i - j) \bmod n \in g$, i.e. the gap $j - i$ lies in $D = \{m\} \cup \{m+2, \dots, 2m\} \subseteq [m, 2m]$; every backedge gap is $\geq m$. A backedge clique $a_1 < \dots < a_k$ has consecutive gaps $\geq m$, so spread $a_k - a_1 \geq (k - 1)m$; as the spread is $\leq 2m$ this forces $k \leq 3$. A backedge triangle has both gaps $= m$ and spread $2m$, hence equals $\{0, m, 2m\}$.

Lower bound $\bar{\omega}(AC_n) \geq 3$. The domination number satisfies $\text{dom} \leq \bar{\omega}$ (Property 3.2 of arXiv:2310.04265). With $N_0 = \{0\} \cup g$, $\text{dom} \leq 2$ iff some translate $t \neq 0$ gives $|N_0 \cap (N_0 + t)| = 1$; writing N_0 as the interval $\{0, \dots, m+1\}$ minus the point m , a short count gives $\min_{t \neq 0} |N_0 \cap (N_0 + t)| = 2$ for all $m \geq 3$ (the autocorrelation lemma; values $m-1, m+1-t, 2$ for $t = 1, 2 \leq t \leq m-1, t = m$). Hence $\text{dom}(AC_n) \geq 3$ and $\bar{\omega}(AC_n) = 3$.

Criticality. By vertex-transitivity delete $v = 0$. The unique triangle $\{0, m, 2m\}$ uses 0, so the identity order is triangle-free on $AC_n - 0$ and $\bar{\omega}(AC_n - 0) \leq 2$; and $AC_n - 0$ contains the directed triangle $1 \rightarrow 2 \rightarrow (m+3) \rightarrow 1$ (gaps $1, m+1, m-1 \in g$), so $\bar{\omega}(AC_n - 0) = 2$. $\square$

Proposition 2 (Substitution lower bound). *For all tournaments S, H , $\bar{\omega}(S[H]) \geq \bar{\omega}(S) + \bar{\omega}(H) - 1$.*

Proof. Fix any order $\prec$ of $S[H]$. Each block $\{s\} \times V(H)$, under the induced order, has a backedge clique of size $\geq \bar{\omega}(H)$. For each block let $\text{rep}(s)$ be its $\prec$ -minimum; ordered by $\prec$ these reps give an order of S with a backedge clique R of size $\geq \bar{\omega}(S)$. Let s_0 be the $\prec$ -maximum of R (its source: $s_0 \rightarrow s$ for all other $s \in R$). Replace $\text{rep}(s_0)$ by a full backedge $\bar{\omega}(H)$ -clique C of block s_0 : every $x \in C$ satisfies $x \succeq \text{rep}(s_0) \succ \text{rep}(s)$ and $s_0 \rightarrow s$, so x beats $\text{rep}(s)$. Thus $C \cup \{\text{rep}(s) : s \in R \setminus \{s_0\}\}$ is a backedge clique of size $\bar{\omega}(S) + \bar{\omega}(H) - 1$. $\square$

2 The theorem

Throughout, $T := AC_n[AC_n]$, with vertices (a, b) , $a, b \in \mathbb{Z}/n$; a is the *outer* and b the *inner* coordinate, and $(a, b) \rightarrow (a', b')$ iff $(a \neq a' \text{ and } (a' - a) \in g)$ or $(a = a' \text{ and } (b' - b) \in g)$. Define the *potential*

$$c(t) = \begin{cases} 3 & t = 0, \\ 2 & 1 \leq t \leq m, \\ 1 & m+1 \leq t \leq 2m. \end{cases}$$

Theorem 1. *For every odd $n \geq 7$, $T = AC_n[AC_n]$ is $5\text{-}\bar{\omega}$ -critical: $\bar{\omega}(T) = 5$ and $\bar{\omega}(T - v) = 4$ for all v . Consequently $\{AC_n[AC_n] : n \geq 7 \text{ odd}\}$ is an infinite family of $5\text{-}\bar{\omega}$ -critical tournaments, proving Conjecture 5.10 at $k = 5$, and Question 5.9 fails at $k = 5$.*

Since T is vertex-transitive $((x, y) \mapsto (x + s, y + t) \text{ are automorphisms})$, it suffices to delete $v = (0, 0)$. The proof occupies the next three sections.

3 Value: $\bar{\omega}(T) = 5$

By Proposition 2 with $S = H = AC_n$ and Proposition 1(a), $\bar{\omega}(T) \geq 3 + 3 - 1 = 5$. For the matching upper bound, order the n^2 vertices by the key $(c(a) + c(b), a, b)$; an elementary casework on the key classes (using only the n -independent arc-facts of Lemma 1) shows the backedge clique number is 5. Combined with $\bar{\omega}(T) \geq 5$ this gives $\bar{\omega}(T) = 5$. Alternatively, $\bar{\omega}(T) \leq 5$ follows from $\bar{\omega}(T - (0, 0)) \leq 4$ (Section 5) since $(0, 0)$ is the $\prec$ -maximum of the order used there: a backedge clique either omits $(0, 0)$ (size ≤ 4) or contains it as its top vertex, the rest being a backedge clique of $T - (0, 0)$ of size ≤ 4 .

4 Deletion, lower bound: $\bar{\omega}(T - (0, 0)) \geq 4$

$T - (0, 0)$ contains the induced subtournament on $\{(a, b) : a \neq 0\}$ (it also retains every $(0, b)$, $b \neq 0$); that induced subtournament is $(AC_n - 0)[AC_n]$. By monotonicity, Proposition 2, and $\bar{\omega}(AC_n - 0) = 2$ (Proposition 1(b)),

$$\bar{\omega}(T - (0, 0)) \geq \bar{\omega}((AC_n - 0)[AC_n]) \geq 2 + 3 - 1 = 4.$$

5 Deletion, upper bound: $\bar{\omega}(T - (0, 0)) \leq 4$

Order the $n^2 - 1$ survivors by **inner-then-outer**: key $(a, b) = (c(b), c(a), a, b)$. The *cell* of a vertex is $\chi(a, b) = (c(b), c(a)) \in \{1, 2, 3\}^2$. As $(0, 0)$ is the only cell- $(3, 3)$ vertex, the survivors occupy the 8 cells $\{1, 2, 3\}^2 \setminus \{(3, 3)\}$.

5.1 One vertex per cell

Lemma 2. *There is no backedge between two vertices of the same cell. Hence a backedge clique has at most one vertex per cell, and so injects into the 8 cells.*

Proof. Fix a cell (β, α) and two of its vertices $(a, b), (a', b')$; then b, b' lie in one length- m monotone interval (or $b = b' = 0$) and likewise a, a' . If $a \neq a'$, the backedge condition between them is $(a_{\text{earlier}} - a_{\text{later}}) \in g$ with both in one length- m interval; by Lemma 1(iii) this residue lies in $[m + 2, 2m]$, disjoint from g . If $a = a'$ and $b \neq b'$, the same applies to the inner gap. Either way there is no backedge. $\square$

5.2 Band caps

Group the 8 cells by inner band $c(b)$: band 1 = $\{(1, 1), (1, 2), (1, 3)\}$, 2 = $\{(2, 1), (2, 2), (2, 3)\}$, 3 = $\{(3, 1), (3, 2)\}$. By Lemma 2, writing $n_j = |S \cap \text{band } j|$, immediately $n_1, n_2 \leq 3$ and $n_3 \leq 2$. (For n_3 : band 3 is $\{(a, 0) : a \neq 0\}$ ordered by $(c(a), a)$ as $[m + 1, \dots, 2m, 1, \dots, m]$; within each half consecutive gaps land in $[m + 2, 2m] \notin g$, so there is no backedge inside a half, giving $n_3 \leq 2$.)

5.3 The in-neighbourhood lemma

Lemma 3 (H17). *For $x \in H$ and $y \in L$, the common in-neighbourhood $N^-(x) \cap N^-(y) = (x - g) \cap (y - g)$ in AC_n lies in a single band: it is contained in $[0, m - 1]$ if $x - y \leq m$, and in $[m + 1, 2m - 1]$ if $x - y \geq m + 1$. In particular it never contains both an H -residue and an L -residue.*

Proof. $N^-(v) = v - g = [v - m - 1, v - 1] \setminus \{v - m\}$. For $x \in \mathbf{H}$ this is a non-wrapping arc; for $y \in \mathbf{L}$ it wraps. Their intersection is $[x - m - 1, y - 1] \subseteq [0, m - 1]$ when $x - y \leq m$, and $[y + m, x - 1] \subseteq [m + 1, 2m - 1]$ when $x - y \geq m + 1$ —single band in either case. $\square$

5.4 No five cells are realizable

The 8 cells in key order are $(1, 1) \prec (1, 2) \prec (1, 3) \prec (2, 1) \prec (2, 2) \prec (2, 3) \prec (3, 1) \prec (3, 2)$. We use a fixed list of 20 **infeasible cell-sets**. Two facts give $\bar{\omega}(T - (0, 0)) \leq 4$:

- (a) each of the 20 sets is infeasible (proved below, n -independently);
- (b) every 5-subset of the 8 cells contains one of the 20—a purely combinatorial statement ($\binom{8}{5} = 56$ checks, independent of n).

By Lemma 2 a backedge clique injects into the 8 cells; by (a)+(b) it cannot occupy 5 of them, so has ≤ 4 vertices. We do *not* claim the list is the exact set of minimal obstructions—only (a),(b), which suffice. We record the *beat conditions*: the higher cell's representative (a_H, b_H) beats the lower's (a_L, b_L) iff $(a_L - a_H) \in g$ when $a_H \neq a_L$, else $(b_L - b_H) \in g$.

(I) Ten triples in five $c(a)=1 \leftrightarrow 2$ mirror pairs. We give the band-H chain of each; the band-L mirror uses Lemma 1(ii) and is written afterwards.

- $\{(1, 1), (1, 3), (2, 1)\} (\cong \{(1, 2), (1, 3), (2, 2)\})$: reps $x \in (1, 1)$, $y = (0, \cdot) \in (1, 3)$, $z \in (2, 1)$. y beats x : $a_x \in g$, $a_x \in \mathbf{H} \Rightarrow a_x = m + 1$. z beats y : $(-a_z) \in g \Rightarrow a_z \in [m + 2, 2m]$. z beats x (same band, $a_z \neq a_x$): $(a_x - a_z) = (m + 1 - a_z) \in g$, but $m + 1 - a_z \in [m + 2, 2m] \notin g$. Contradiction.
- $\{(1, 1), (1, 3), (3, 1)\} (\cong \{(1, 2), (1, 3), (3, 2)\})$: same chain with $z \in (3, 1)$ ($a_x = m + 1$, $a_z \in [m + 2, 2m]$, z beats x needs $(m + 1 - a_z) \in g$, false).
- $\{(1, 1), (2, 3), (3, 1)\} (\cong \{(1, 2), (2, 3), (3, 2)\})$: $y \in (2, 3)$, $z \in (3, 1)$; identical ($a_x = m + 1$, $a_z \in [m + 2, 2m]$, false).
- $\{(2, 1), (2, 3), (3, 1)\} (\cong \{(2, 2), (2, 3), (3, 2)\})$: $x \in (2, 1)$, $a_x = m + 1$; $z \in (3, 1)$, $a_z \in [m + 2, 2m]$; z beats x needs $(m + 1 - a_z) \in g$, false.
- $\{(1, 3), (2, 1), (2, 3)\} (\cong \{(1, 3), (2, 2), (2, 3)\})$: reps $u = (0, \cdot) \in (1, 3)$, $v \in (2, 1)$, $w = (0, \cdot) \in (2, 3)$. v beats u : $(-a_v) \in g \Rightarrow a_v \in [m + 2, 2m]$. w beats v : $a_v \in g \Rightarrow a_v = m + 1$, contradicting $a_v \in [m + 2, 2m]$.

Band-L mirrors. The first four: $y = (0, \cdot)$ beats $x = (a_x, \cdot)$ forces $a_x \in g$, $a_x \in \mathbf{L} \Rightarrow a_x \in [1, m - 1]$; the beaten-by-source coordinate a_z satisfies $(-a_z) \in g$, $a_z \in \mathbf{L} \Rightarrow a_z = m$; then z beats x needs $(a_x - m) \equiv a_x + m + 1 \in [m + 2, 2m] \notin g$. The fifth: $v \in (2, 2)$ gives $(-a_v) \in g \Rightarrow a_v = m$, and $w = (0, \cdot)$ beats v needs $a_v \in g$, but $a_v = m \notin g$. All contradictions.

(II) Four outer-source quadruples (cells in key order; every step is an arc-fact of Lemma 1):

- $\{(1, 1), (1, 2), (2, 1), (2, 3)\}$: $p \in (1, 1)$, $q \in (1, 2)$, $r \in (2, 1)$, $s = (0, \cdot) \in (2, 3)$ (top). s beats $p, r \Rightarrow a_p = a_r = m + 1$ (same block). Then $q \succ p$ needs $(m + 1 - a_q) \in g$ while $r \succ q$ needs $(a_q - (m + 1)) \in g$: opposite arcs between blocks a_q and $m + 1$, impossible.
- $\{(1, 2), (2, 1), (2, 2), (2, 3)\}$: $p \in (1, 2)$, $q \in (2, 1)$, $r \in (2, 2)$, $s = (0, \cdot) \in (2, 3)$. s beats $q \Rightarrow a_q = m + 1$; s beats $p, r \Rightarrow a_p, a_r \in [1, m - 1]$. $q \succ p$: $(a_p - (m + 1)) \equiv a_p + m \in g \Rightarrow a_p = 1$. $r \succ q$: $(m + 1 - a_r) \in g \Rightarrow a_r \in [2, m]$. $r \succ p$ (both in $\mathbf{L}$, $a_r \neq 1$): $(1 - a_r) \equiv 2m + 2 - a_r \in [m + 2, 2m] \notin g$.

- $\{(1, 3), (2, 1), (2, 2), (3, 1)\}$: $p = (0, \cdot) \in (1, 3)$ (bottom), $q \in (2, 1), r \in (2, 2), s = (a_s, 0) \in (3, 1)$. Each of q, r, s beats p : $a_q \in [m + 2, 2m]$, $a_r = m$, $a_s \in [m + 2, 2m]$. $r \succ q$: $(a_q - m) \in g \Rightarrow a_q \in [m + 2, 2m - 1]$. $s \succ r$: $(m - a_s) \in g \Rightarrow a_s = 2m$. $s \succ q$ (both in H): $(a_q - 2m) \equiv a_q + 1 \in [m + 3, 2m] \notin g$.
- $\{(1, 3), (2, 2), (3, 1), (3, 2)\}$: $p = (0, \cdot) \in (1, 3)$, $q \in (2, 2), r = (a_r, 0) \in (3, 1), s = (a_s, 0) \in (3, 2)$. Each beats p : $a_q = m$, $a_r \in [m + 2, 2m]$, $a_s = m$ (q, s share block m). $r \succ q$: $(m - a_r) \in g \Rightarrow a_r = 2m$. $s \succ r$: $(a_r - a_s) = 2m - m = m \notin g$.

(III) Six squares

$$\begin{aligned} &\{(1, 1), (1, 2), (2, 1), (2, 2)\}, \{(1, 1), (1, 2), (2, 1), (3, 2)\}, \{(1, 1), (1, 2), (3, 1), (3, 2)\}, \\ &\{(1, 1), (2, 2), (3, 1), (3, 2)\}, \{(1, 2), (2, 1), (2, 2), (3, 1)\}, \{(2, 1), (2, 2), (3, 1), (3, 2)\}. \end{aligned}$$

In key order each has $c(a)$ alternating H, L, H, L : two cells in band H , two in band L . Write the two band- H reps and two band- L reps; the two same-band cell-pairs branch:

- *equality branch* (two same-band reps share a block): two of the six backedges become opposite arcs $x \rightarrow y, y \rightarrow x$ between the shared block and a third block—impossible by tournament antisymmetry.
- *all-distinct branch*: the six outer backedges force one band- H rep and one band- L rep to be common in-neighbours, one in each band, of the other band- H and band- L rep—impossible by Lemma 3.

Both branches are contradictory for each of the six squares.

This establishes (a) for all 20 sets; with (b) we conclude $\bar{\omega}(T - (0, 0)) \leq 4$. $\square$

6 Conclusion

By Sections 4–5, $\bar{\omega}(T - (0, 0)) = 4$; by vertex-transitivity $\bar{\omega}(T - v) = 4$ for every v . With $\bar{\omega}(T) = 5$ (Section 3), $T = AC_n[AC_n]$ is 5- $\bar{\omega}$ -critical for every odd $n \geq 7$. Distinct n give tournaments of distinct order n^2 , so this is an infinite family, proving **Conjecture 5.10 at $k = 5$** .

Question 5.9. A proper subtournament $A \subsetneq T$ omits a vertex v , so $A \subseteq T - v$ and $\bar{\omega}(A) \leq \bar{\omega}(T - v) = 4$ by monotonicity. Thus the only subtournament of T with $\bar{\omega} \geq 5$ is T itself, of unbounded order n^2 ; no bound $\ell(5)$ exists, i.e. **Question 5.9 fails at $k = 5$** . $\square$

Verification. Every arc-fact used in Section 5 is n -independent and was checked for $m = 3, \dots, 200$; the 20-set list, the coverage (b), and Lemma 3 were checked exactly for small odd n . The construction is reproducible with the exact $\bar{\omega}$ oracle (exact branch-and-bound and a validated no- K -clique SAT encoding). An independent multi-skeptic red-team reproduced the value, the deletion, $\bar{\omega}(AC_n - 0) = 2$, and Lemma 3.